Rare Mendelian Disease Diagnosis via Trio-Based Genomic Analysis of Simulated Exome Data

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Abstract :

The molecular diagnosis of rare Mendelian disorders requires optimal computational workflows to identify causative variants within vast genomic datasets. This study aimed to reproduce a realistic diagnostic pipeline using simulated whole-exome sequencing (WES) data from five family trios, derived from the 1000 Genomes Project backgrounds. Focusing exclusively on human chromosome 20 (GRCh38), potential pathogenic variants were investigated following specific inheritance models: autosomal recessive, inherited autosomal dominant, and de novo autosomal dominant. Using a UNIX-based bioinformatics pipeline and Ensembl's Variant Effect Predictor (VEP), different pathogenic variants were prioritized based on population frequency and predicted impact on gene function. The analysis successfully identified candidate causative variants and clinical-associated phenotypes, demonstrating the efficacy of integrating inheritance-based filtering with computational prioritization. This project underscores the importance of standardized genomic workflows in clinical diagnostics for the identification of rare genetic diseases.

Index Terms: Variant Calling, Rare Disease

1 Introduction

The molecular diagnosis of rare genetic disorders has been significantly advanced by the integration of sequencing techniques, especially with the latest methods, such as Whole Exome Sequencing (WES), that focuses on exome regions, where the majority of known disease-causing variants reside. However, establishing a definitive diagnosis remains a challenge and a precise bioinformatic approach is required to distinguish pathogenic variants from the vast background of neutral variation. The effectiveness of the Exome Sequencing depends on robust reference frameworks provided by large-scale population studies. The 1000 Genomes Project established a critical baseline by providing a validated haplotype map of millions of variants across diverse populations. This resource confirms that low-frequency variants are significantly enriched for deleterious mutations.

In this case-study, a family trio analysis was employed to leverage Mendelian segregation patterns. By sequencing the child alongside both biological parents, the diagnostic pipeline effectively prioritizes variants based on the specific inheritance mode - Autosomal Recessive (AR), Autosomal Dominant (AD) inherited, or AD de novo — assigned to each case. In fact, molecular diagnosis requires that the variants interpretation must be contextualized within an individual's local genetic background. This work details a complete diagnostic workflow applied to five family trios, focusing on human chromosome 20 in the GRCh38 (hg38) reference genome. It is important to note that the causative variants in this cohort are simulated, although they are embedded within real genetic backgrounds derived from the 1000 Genomes Project.

The pipeline utilizes Bowtie2 for read mapping, Freebayes for haplotype-based variant calling, and the Ensembl Variant Effect Predictor (VEP) for functional annotation. Variants are prioritized using known impact, allele frequency and pathogenicity scores, specifically SIFT and PolyPhen-2, to identify the causative variant responsible for the associated clinical phenotype in each trio.

1.1 Background

The 1000 Genomes Project was started to create a complete public list of common human genetic variations, specifically those found in at least 1% of the populations studied. With the advantage of the rapid progress in sequencing technology that significantly lowered costs, it became the first project to sequence a massive number of individual genomes, eventually including 2,504 people from 26 different populations[1]. This major effort went through several phases and finished its main publication in 2015. It was essential for identifying millions of SNPs, short indels, and structural variants. Using real genetic backgrounds from the 1000 Genomes Project allowed to simulate a realistic diagnostic setting while remaining ethically safe, as it avoids the privacy risks associated with using sensitive patient data. The guaranteed presence of disease-causing variants in the dataset also allowed to identify them even if the sequencing coverage was not perfect.

2 Materials and Methods

2.1 Data Acquisition, Storage and Genomic Context

The analysis was performed on genomic data from five family trios (a child and both biological parents). The datasets were simulated using real genetic backgrounds provided by the 1000 Genomes Project in order to perform the analysis. For this reason, all data, all the intermediate results and files, including the reference Chr20 genome can be found on the lab server leon, 159.149.160.7, in the directory ~BCG2026_Marsigliani_T/exam_project_RESULTS, while input and final sumamry can also be found in the project repository[2] via GitHub. While the sequencing data remain simulated, the reference genome of human chromosome 20 follows the GRCh38 annotation. Because this study aims to prioritize variants for diagnosing rare Mendelian disorders, a specific mode of inheritance was also provided for each family trio (Table 1). In addition to the inheritance models table, details for the individuals associated with each genotype (child, father, and mother) were provided:

HG00445 - Child; HG00446 - Father; HG00447 - Mother.

Trio	Inheritance Mode	Notes
trio_1	Autosomal Recessive	NA
trio_2	Autosomal Dominant inherited	Mother affected
trio_3	Autosomal Dominant - de novo	NA
trio_4	Autosomal Recessive	NA
trio_5	Autosomal Recessive	NA

Table 1. Summary of inheritance models for the five family trios.

2.2 Data Analysis Pipeline

To ensure a standardized and reproducible analysis across the entire cohort, each of the following processing modules was executed iteratively for every family trio through a for loop that automated folder navigation and sample selection using the following command structure:

```
for trio in trio_1 trio_2 trio_3 trio_4
    trio_5; do
echo "===␣WORKING␣ON␣$trio␣==="
  cd $trio
```

For each member of the family, raw paired-end reads in FASTQ format were aligned to the reference genome using Bowtie2, a fast and memory-efficient tool that leverages a genome index to align sample reads to the reference sequence. To optimize the workflow, a Bowtie2-pre-built index of human chromosome 20 (GRCh38) provided in the project's shared directory was used, avoiding the need for a preliminary indexing step. Furthermore, to ensure downstream reproducibility and proper sample identification, read group tags (–rg-id and –rg SM) were assigned to each family member during alignment. The resulting SAM files were directly piped into SAMtools for conversion to binary format (BAM), coordinate sorting and indexing. The SAM/BAM format, together with SAMtools, allow the separation of the alignment step from downstream analyses, enabling a generic and modular approach to the analysis of genomic sequencing data[3].

```
echo "---␣Aligning␣reads␣---"
bowtie2 -1 HG00445.targets_R1.fq.gz -2
    HG00445.targets_R2.fq.gz -x /home/
    BCG2026_exam/chr20 --rg-id "child" --rg
    "SM:child" | samtools view -Sb |
    samtools sort -o ${trio}_child.bam
```

```
echo "---␣Indexing␣BAMs␣---"
samtools index ${trio}_child.bam
```

A dual-stage quality control strategy was implemented: sequencing quality was assessed with FastQC to evaluate per-base quality scores; post-alignment metrics, including mean coverage and depth distribution across the targeted exonic regions, were calculated using Qualimap for each member of the family. Like many bioinformatics programs, FastQC and Qualimap[4] produce logs detailing their results that are excellent at highlighting potential problems in data. However, nearly all of these logs and reports are produced on a per-sample basis, requiring the user to find and compile multiple QC results. This process is time consuming, repetitive and complex, making it prone to errors. MultiQC addresses this problem by scanning given analysis directories for log files and QC reports, creating a single summary report visualizing results across all samples. Collecting data within a single report provides a fast way to scan key statistics quickly and easily. Shared plots allow accurate comparison between samples, allowing detection of subtle differences not noticeable when switching between different files[5]. For this reason, all QC reports were aggregated after the for loop end, into a single interactive summary using MultiQC to ensure data consistency across the cohort.

```
echo "---␣Running␣FastQC␣---"
fastqc *.bam
```

```
echo "---␣Running␣Qualimap␣---"
qualimap bamqc -bam ${trio}_child.bam --
    feature-file ../chr20_ILMN_Exome_2.0
    _Plus_Panel.hg38_padded.bed -outdir ${
    trio}_child
```

At this point, the direct detection of haplotypes from alignment data presents several challenges to some variant detection methods. Most variant detection methods establish estimates of the likelihood of polymorphism at a given loci using statistical models which only assume biallelism and uniform, typically diploid, copy number. This issue, in parallel with improper modeling of copy number impedes the really accurate detection of small variants on sex chromosomes, in polyploid organisms, or in locations with known copy-number variations, where called alleles, genotypes, and likelihoods should reflect local copy number and global ploidy. To enable the application of population-level inference methods to the detection of haplotypes and in order to solve the cited problems above, freebayes generalize the Bayesian statistical method[6] to allow multiallelic loci and non-uniform copy number across the samples under consideration[7]. For this reason, the variant calling step was performed for each trio using Freebayes, obtaining for each trio a Variant Call Format file (${trio}.vcf). To minimize false positives, stringent filters were applied during the calling process: a minimum mapping quality of 20 (-m 20), at least 5 supporting alternate reads (-C 5), a base quality threshold of 10 (-Q 10), and a minimum total coverage of 10x.

```
echo "---␣Variant␣calling␣---"
freebayes -f /home/BCG2026_exam/chr20.fa -m
    20 -C 5 -Q 10 --min-coverage 10 ${trio}
    _child.bam ${trio}_father.bam ${trio}
    _mother.bam > ${trio}.vcf
```

Variants candidates were subsequently filtered, using the BCFtools filtering commands, based on the provided mode of inheritance for each case (a local variable "FILTER" was actually used) in the table above (Table 1):

- Autosomal Recessive (AR): Variants where the child was homozygous for the alternate allele (AA) while both parents were heterozygous (AR).
- Autosomal Dominant (AD) Inherited: Variants where the child and the affected parent shared a heterozygous genotype (AR), with the healthy parent being homozygous reference (RR).
- AD de novo: Variants where the child was heterozygous (AR) and both parents were homozygous reference (RR).

```
bgzip ${trio}.vcf
bcftools index ${trio}.vcf.gz
```

```
if [ $trio == "trio_1" ] || [ $trio == "
    trio_4" ] || [ $trio == "trio_5" ]; then
    FILTER='GT[0]="AA"␣&&␣GT[1]="RA"␣&&␣GT
        [2]="RA"'
elif [ $trio == "trio_2" ]; then
    FILTER='GT[0]="RA"␣&&␣GT[1]="RR"␣&&␣GT
        [2]="RA"'
elif [ $trio == "trio_3" ]; then
    FILTER='GT[0]="RA"␣&&␣GT[1]="RR"␣&&␣GT
        [2]="RR"'
fi
```

To optimize computational efficiency and focus on relevant regions, calling was restricted to the target panel using the provided BED file (also available in the related repo).

```
echo "---␣Filtering␣variants␣---"
bcftools view -R ../chr20_ILMN_Exome_2.0
    _Plus_Panel.hg38_padded.bed ${trio}.vcf.
    gz | bcftools view -S ../samples.txt |
    bcftools view -i "${FILTER}" | bcftools
    filter -i 'QUAL>20' -Ov -o ${trio}.cand.
    vcf
```

Candidate variants were annotated using the Ensembl Variant Effect Predictor (VEP). The VEP is a software that performs annotation and analysis of many types of variation in coding and non-coding regions of the genome. It is a critical tool to annotate variants and prioritize a subset for further analysis[8]. Prioritization was based on: Predicted Impact - focus on variants with "HIGH" or "MODERATE" impact; Allele Frequency - filtering against the 1000 Genomes and gnomAD databases, retaining only rare variants with a allele frequency (AF) under 0.0001; Pathogenicity Scores - In silico predictions from SIFT and PolyPhen-2 were utilized to assess the likelihood of protein-function disruption too.

```
echo "-␣Annotating␣and␣Filtering␣with␣VEP␣-"
vep -i ${trio}.cand.vcf -o ${trio}.
    vep_annotated.vcf --vcf --cache --
    offline --assembly GRCh38 --dir_cache /
    data/vep_cache --use_given_ref --mane --
    pick_allele --af --af_1kg --af_gnomade
    --max_af --sift b --polyphen b --
    no_fasta
filter_vep -i ${trio}.vep_annotated.vcf -o $
    {trio}.vep_filtered.vcf --filter "(
    IMPACT␣is␣HIGH␣or␣IMPACT␣is␣MODERATE)␣
    and␣(not␣MAX_AF␣or␣MAX_AF␣<␣0.0001)"
```

As last step, coverage tracks for each family member and trio were generated with Bedtools genomecov, capped at 100x depth, and converted to BedGraph format for visualization of the causative loci track using the UCSC Genome Browser.

```
echo "---␣Computing␣coverage␣tracks␣---"
bedtools genomecov -ibam ${trio}_father.
    bam -bg -trackline -trackopts 'name="
    father"' -max 100 > ${trio}_fatherCov.
    bg
```

Furthermore, as said before, a command to aggregate all QC reports into a single interactive MultiQC report was performed.

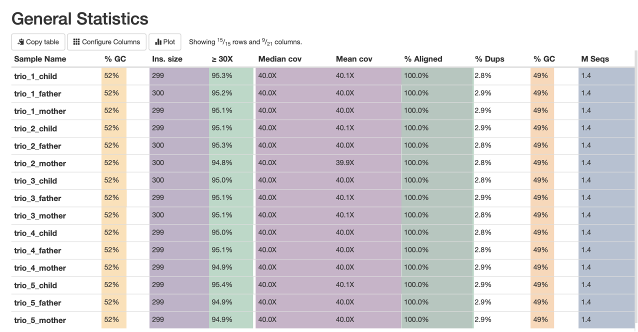
```
echo "---␣Running␣MultiQC␣---"
multiqc trio_*/
```

2.3 *Variant Effect Predictor online tool validation*

To ensure a more reliable and verified result, the candidates identified by the filter_vep command-line tool were further investigated using the official VEP online platform[9] (GRCh38 version) by uploading the files containing the candidate variants. This step was performed because the online version handles indels more robustly and is able to match them to associated phenotypes even when their coordinates do not exactly correspond to those reported in the Variant Call Format file. Such discrepancies can arise because different variant calling tools may represent indels in slightly different ways. For this reason, online results were preferred for diagnosis.

3 Results

The initial quality assessment was performed on FASTQ files and aggregated via MultiQC. The General Statistics (Figure 1) demonstrate high data consistency across all five family trios. Each sample yielded approximately 1.4 million reads, with an alignment rate of 100.0% to the GRCh38 reference genome. The mean coverage across the targeted regions was recorded at 40x (ranging from 39.9x to 40.1x), with over 95% of the target exome covered at more than 30x.


General Statistics

Copy table | Configure Columns | Plot | Showing 15/15 rows and 9/21 columns.

Sample Name	% GC	Ins. size	≥ 30X	Median cov	Mean cov	% Aligned	% Dups	% GC	M Seqs
trio_1_child	52%	299	95.3%	40.0X	40.1X	100.0%	2.8%	49%	1.4
trio_1_father	52%	300	95.2%	40.0X	40.0X	100.0%	2.9%	49%	1.4
trio_1_mother	52%	299	95.1%	40.0X	40.0X	100.0%	2.9%	49%	1.4
trio_2_child	52%	299	95.1%	40.0X	40.1X	100.0%	2.9%	49%	1.4
trio_2_father	52%	300	95.3%	40.0X	40.0X	100.0%	2.8%	49%	1.4
trio_2_mother	52%	300	94.8%	40.0X	39.9X	100.0%	2.8%	49%	1.4
trio_3_child	52%	300	95.0%	40.0X	40.0X	100.0%	2.8%	49%	1.4
trio_3_father	52%	299	95.1%	40.0X	40.1X	100.0%	2.9%	49%	1.4
trio_3_mother	52%	300	95.1%	40.0X	40.1X	100.0%	2.8%	49%	1.4
trio_4_child	52%	299	95.0%	40.0X	40.0X	100.0%	2.8%	49%	1.4
trio_4_father	52%	299	95.1%	40.0X	40.0X	100.0%	2.8%	49%	1.4
trio_4_mother	52%	299	94.9%	40.0X	40.0X	100.0%	2.9%	49%	1.4
trio_5_child	52%	299	95.4%	40.0X	40.1X	100.0%	2.9%	49%	1.4
trio_5_father	52%	299	94.9%	40.0X	40.0X	100.0%	2.9%	49%	1.4
trio_5_mother	52%	299	94.9%	40.0X	40.0X	100.0%	2.9%	49%	1.4

Figure 1. MultiQC General Statistics summary

The sequence quality was further validated by the Per Base Sequence Quality plot and Per Sequence Quality plot (Figure 2 and 3), which show that the mean PHRED scores remain well above 30 (99.9% accuracy) across the entire 150 bp read length.

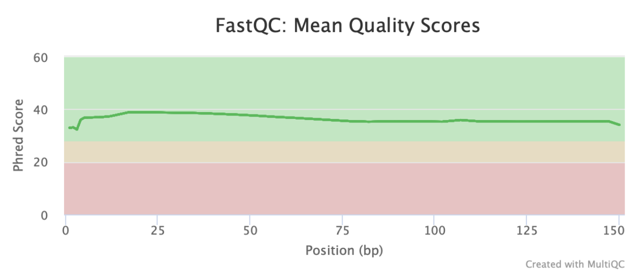


Figure 2. FastQC: Mean quality scores (PHRED) per position.

The Qualimap Coverage Histogram (Figure 4) confirms a robust enrichment of reads within the target regions. The peak of the

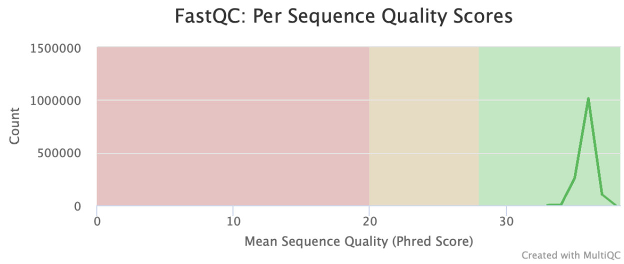


Figure 3. FastQC: Per sequence quality score.

distribution aligned with the calculated mean of 40x, indicating a uniform sequencing process across all trios without significant gaps in coverage that could hinder the diagnostic process.

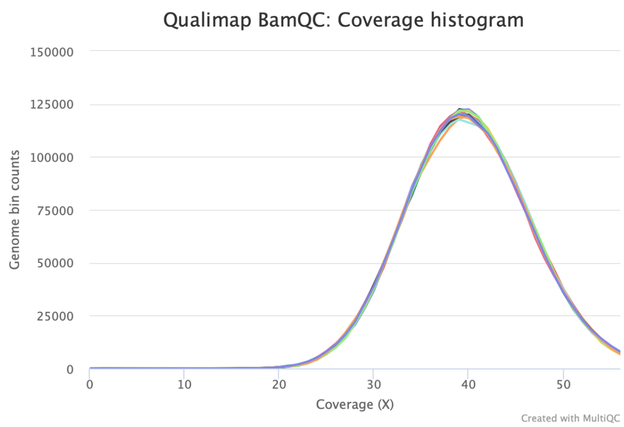


Figure 4. Genome fraction coverage histogram.

Joint variant calling with Freebayes identified many different genetic variations. By applying inheritance (AR, AD inherited, AD de novo) and absolute frequence (AF < 0.0001) filters, the following causative variants in Table 2 were identified for each trio (also a double-check with pathogenicity scores - SIFT and PolyPhen-2 - has been executed). The causative variants, togheter with their trio coverage tracks were manually inspected using the UCSC Genome Browser[10]. Figure 5 illustrates a representative example of a causative locus on Trio 4, showing the alignment tracks for the child and parents, confirming the presence of the mutation (C//T) in the affected child.

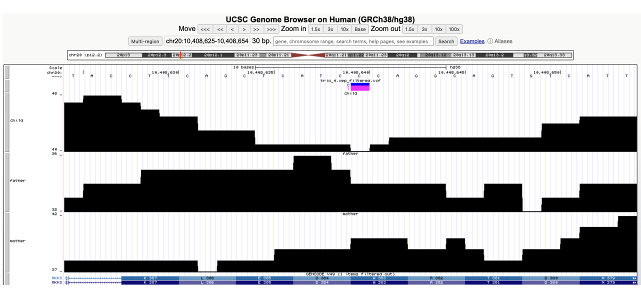


Figure 5. Bedgraph track visualization via UCSC Genome Browser for Trio 4 (AR Inheritance) with the filtered mutation on the gene MKKS.

4 Discussion

Before starting the discussion of the results, it is necessary to specify that the exceptional uniformity in sequencing depth and the perfect alignment rates observed across all samples are a direct consequence of the simulated nature of the datasets, which were computationally generated using real genetic backgrounds from the 1000 Genomes Project to ensure a controlled, reproducible, and standardized environment for the diagnostic pipeline evaluation.

The primary objective of this diagnostic workflow was to isolate highly likely pathogenic candidates from the vast background of neutral human variation. By leveraging Mendelian segregation patterns and focusing on high-impact variants within chromosome 20, the pipeline successfully identified the molecular basis for clinical pathogenic phenotypes reported in the Table 2.

4.1 Trio 1

The analysis of the first trio, conducted under the assumption of an autosomal recessive (AR) inheritance model, yielded no clinically significant variants, both through the CLI and the online VEP tool. Despite the high sequencing depth and data quality (consistent with the simulated nature of the project), no candidate variants passed the filtering criteria regarding allele frequency and functional impact. Consequently, this case was classified as healthy for pathogenic variants on human chromosome 20.

4.2 Trio 2

In the second trio, a heterozygous frameshift variant was identified in the SAMHD1 gene. The mutation follows an autosomal dominant (AD) inheritance pattern, specifically inherited from the mother. This genetic finding is consistent with a diagnosis of Aicardi-Goutières syndrome, a severe neuroinflammatory disorder characterized by early-onset encephalopathy and autoimmune features[11]. A special mention is necessary for this trio: this specific case reported two different results from the CLI command and the VEP online tool. Since the local CLI tool reflects an older update compared to the online version, the SAMHD1 Frameshift variant couldn't be identified and the research continued till the second result (also identified by the VEP online tool), a Moderate-impact Inframe deletion in the MYT1 gene, related to the *MYT1-related oculo-auriculo-vertebral spectrum OAVS Goldenhar syndrome* caused by the mutation AGAGGAGGAGGAGGAGGAT // AGAGGAGGAGGAGGAT in the position 20:64208000-64208018.

4.3 Trio 3

The analysis of the third trio revealed a heterozygous de novo frameshift variant in the JAG1 gene. The de novo nature of this event is confirmed by its presence in the child and its absence in both healthy biological parents. This mutation is causative for Alagille syndrome, a multisystem disorder primarily affecting the liver, heart, and skeleton[12]. The identification of a high-impact mutation in JAG1 provides a definitive molecular diagnosis, due to the loss-of-function variants effect in this gene.

4.4 Trio 4

The fourth trio follows an autosomal recessive inheritance model. A homozygous stop-gained mutation has been identified in the MKKS gene. This nonsense mutation results in a truncated pro-

Trio	Inheritance	Mutated Gene	Locus	Mutation	Consequence	Impact	Associated phenotype
1	AR	NA	NA	NA	NA	NA	Healthy
2	AD (mother affected)	SAMHD1	20:36898463-36898465	CTC/CC	Frameshift variant	High	Aicardi-Goutieres syndrome
3	AD (de novo)	JAG1	20:10639939-10639943	CAAGG/ CAAAGG	Frameshift variant	High	Alagille syndrome due to a JAG1 point mutation
4	AR	MKKS	20:10408640-10408640	C/T	Stop gained	High	Bardet-Biedl syndrome
5	AR	ADA	20:44621017-44621017	C/T	Splice donor variant	High	Severe combined immunodeficiency autosomal recessive T cell-negative, B cell-negative, NK cell-negative due to ADA deficiency

Table 2. Identified rare phatogenic variants for trios with related clinical phenotypes validated via VEP online tool.

tein and is the molecular cause of Bardet-Biedl syndrome, a complex ciliopathy characterized by obesity, polydactyly, and renal anomalies[13]. The identification of a premature termination codon aligns with the loss-of-function mechanism typically associated with this recessive phenotype.

4.5 Trio 5

Also in the final trio as in the second one, two different results arise. An autosomal recessive inheritance pattern led to the identification of a homozygous splice donor variant in the ADA gene from the VEP online tool. This variant disrupts the highly conserved 2-base region at the beginning of an intron, leading to aberrant splicing and protein dysfunction. This mutation is the cause of Severe Combined Immunodeficiency (SCID) due to Adenosine Deaminase deficiency. This condition is characterized by a lack of functional T, B, and NK cells, leading to severe immune failure in infancy[14]. Conversely, the CLI couldn't find any variants reflecting the filtering criteria set and no results were given.

5 Conclusions

In conclusion, the bioinformatic pipeline proposed in this study proved to be highly effective in identifying the molecular causes of rare Mendelian disorders using trio-based exome sequencing data. By focusing the analysis on human chromosome 20 and leveraging the genetic backgrounds provided by the 1000 Genomes Project, the causative variants for each trio were successfully identified. The trio-based approach with the specified inheritance model was fundamental in filtering out the vast majority of neutral genetic variation, allowing for a clear distinction between de novo mutations and inherited pathogenic alleles. The following prioritization strategy, which targeted high-impact variants with low allele frequencies, ensured an important diagnostic yield. While the datasets used were simulated, the exceptional consistency between sequencing metrics and biological findings emphasizes the reliability of this workflow also for clinical applications. It is important to note that potential discrepancies observed between the Ensembl VEP online tool and the CLI version are primarily due to database versioning. The online tool is consistently updated to the latest Ensembl release, whereas the CLI version often utilizes a older annotation version or population frequencies; these differences highlight the necessity of manual verification via the web interface to ensure no significant variants are identified. However, this study also reinforces the role of Whole Exome Sequencing (WES) as an indispensable tool in modern genetics, providing a robust framework for resolving clinical cases and enabling precise genetic counseling.

6 Bibliography

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